

Unit-1

1. Explain why software project management is known as a structured approach.
2. Explain the concept of a software project and the importance of software project management in today's IT industry.
3. Discuss the various activities involved in software project management.
4. What are the key elements in setting objectives, and how do well-defined objectives influence the success of a software project?
5. Compare and contrast different software project management methodologies. How do methods and methodologies impact project planning and execution?
6. Discuss aspects and factors that determine project success or failure
7. Explain the different stages of the Software Project Life Cycle. How does each stage contribute to the successful development and delivery of a software product?
8. Explain various project selection methodologies in software engineering. How do these methods help organizations prioritize and choose projects with maximum business value and feasibility?
9. Explain project selection methodologies like cost-benefit analysis, scoring models, and portfolio management that guide organizations in making strategic investment decisions.
10. Compare and contrast different software process models such as Waterfall, Spiral, Incremental, and Agile. How does the choice of a process model impact the quality and success of a software project?
11. Evaluate the criteria for selecting an appropriate software process model for a given project.
12. What factors should a project manager consider before choosing a life cycle model, and how do these decisions influence the outcome?

Unit-2

1. Explain how Gantt charts, PERT/CPM, SWOT analysis, and work breakdown structures (WBS) assist in effective project planning.
2. Explain how funds flow analysis helps in identifying the financial health of a project, budgeting, and financial decision-making.

3. Explain cost-benefit analysis and its role in software project selection and prioritization.
4. What is risk evaluation in software project management?
5. How does portfolio management contribute to strategic alignment of software projects?
6. Define benefit management and discuss its importance in software project success.
7. Explain the techniques used in gathering requirements during the initial stages of a software project.
8. Describe the process of identifying project scope and preparing a scope statement.
9. Explain the step-wise project planning process proposed by software engineering standards.
10. Discuss various software effort estimation techniques: expert judgment, analogy-based estimation, parametric models like COCOMO and machine learning-based estimation methods.
11. Discuss the impact of schedule compression on software project cost, quality, and risk.
12. Explain the process and challenges involved in software cost estimation: direct vs. indirect costs, fixed vs. variable costs, and internal vs. external costs.

Unit-3

1. Explain how schedule risks are evaluated and mitigated. Include examples of common risks like resource unavailability, scope creep, or underestimated timelines.
2. What are the key components of a risk management plan in software project management?
3. Explain probability-impact matrices, risk exposure, and risk prioritization. Include examples of how risks are ranked and handled based on severity.
4. Describe various techniques used for identifying risks in software project management. Include brainstorming, checklists, expert judgment, SWOT analysis, and historical data.
5. Explain the process of risk management in software projects. Provide examples from real or hypothetical projects.

6. Describe how the integration of WBS, scheduling, and network techniques supports successful project planning and execution.
7. Describe how critical activities influence the overall schedule and how project managers monitor them. Provide an example using a network diagram.
8. Define float in project scheduling. What is the significance of activity float and how is it calculated?
9. Explain how to identify the critical path, calculate activity float, and determine critical activities. Use an example network diagram to illustrate the process.
10. Explain how Gantt charts are used for planning, tracking progress, and managing time. Include an example of a Gantt chart representing a software development life cycle.
11. Describe the steps involved in developing a project schedule in software project management. Include the role of dependencies, durations, resource constraints, and milestones.
12. Differentiate between WBS, Product Breakdown Structure (PBS), and Resource Breakdown Structure (RBS). Illustrate with diagrams how each breakdown structure aids in planning and control of a software project.

Unit-4

1. Describe the importance of contract management in software project success. What are the key steps involved?
2. Explain the process of contract placement in managing software projects.
3. Discuss different types of contracts used in software project management. Give suitable examples.
4. Explain project audit and project reviews. How do they contribute to successful project completion?
5. What is the purpose of status review meetings in project monitoring? Discuss the benefits and challenges.
6. Describe the process of change control in software project management. Why is it important?
7. What is Earned Value Analysis (EVA)? Explain its role in project performance evaluation with an example.

8. Discuss cost control tools used in project monitoring and explain how they help in budget management.
9. Differentiate between performance control and schedule control in project monitoring. Provide examples.
10. Explain progress control in software projects. How can it be effectively implemented?
11. Discuss the criteria and techniques used to prioritize monitoring activities in software projects.
12. Explain the design of a project monitoring system. How does it help in effective project management?

Unit-5

1. Explain the role of leadership and people management skills in handling software project teams.
2. Discuss the essential skills and competencies required by a software project manager. Give practical examples.
3. How does effective communication and coordination improve teamwork in software projects?
4. Explain the challenges of working in teams in a software environment. Suggest measures to overcome them.
5. Discuss the role of project management procedures in ensuring the success of software projects.
6. How do coordination and procedures support effective project management in a software environment?
7. Explain the importance of organizational behaviour in managing software project teams.
8. Discuss various team structures used in software development projects. Provide examples of when each is suitable.
9. Describe projectized organization structure. How does it influence authority and decision making?
10. What is a matrix organization? Discuss its features, benefits, and challenges in software project execution.
11. Explain the advantages and disadvantages of functional organization for managing software projects.

12. Compare and contrast functional, matrix, and projectized organizational structures in software project management.